UC Berkeley

Department of Electrical Engineering and Computer Sciences

EECS 127 - EECS 227AT. Optimization Models in Engineering

Lecture Note 17

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Let's get back to convex optimization:

$$\begin{aligned} \min_{\boldsymbol{x}\in\mathbb{R}^n} \quad & \underbrace{\boxed{f_0(\boldsymbol{x})}}_{\text{convex}}, \\ \text{s.t.} \quad & \underbrace{\boxed{f_i(\boldsymbol{x})}}_{\text{convex}} \le 0,\ i=1,\dots,m, \\ & \underbrace{\boxed{h_j(\boldsymbol{x})}}_{\text{affine}} = 0,\ j=1,\dots,k \end{aligned}$$

under either of these conditions:

(i) f_0 is coercive;

(ii) The feasible set is compact;

then, the set of solutions $\mathcal{X}^{\text{opt}}$ is non-empty (solution exists).
When does $\mathcal{X}^{\text{opt}}$ include only one point?

Theorem 1. *If $f_0(\boldsymbol{x})$ is strictly convex, then there is either no solution or a unique solution.*

For example, $\min \mathrm{e}^x$ has no solution, and $\min x^2$ has a unique solution.

Proof. Assume there are at least two solutions $\boldsymbol{x}^*$ and $\boldsymbol{y}^*$. Consider $\frac{1}{2}\boldsymbol{x}^* + \frac{1}{2}\boldsymbol{y}^*$, which is a convex combination of $\boldsymbol{x}^*$ and $\boldsymbol{y}^*$. Due to convexity of the feasible set, $\frac{1}{2}\boldsymbol{x}^* + \frac{1}{2}\boldsymbol{y}^*$ is also feasible. Also, due to strict convexity,

$$f_0\left(\frac{1}{2}\boldsymbol{x}^* + \frac{1}{2}\boldsymbol{y}^*\right) < \frac{1}{2}f_0(\boldsymbol{x}^*) + \frac{1}{2}f_0(\boldsymbol{y}^*) = f_0(\boldsymbol{x}^*).$$

(The last step is because $\boldsymbol{x}^*$ and $\boldsymbol{y}^*$ are both global minima, so $f_0(\boldsymbol{x}^*) = f_0(\boldsymbol{y}^*)$ is the global optimal objective value.) Since $f_0(\frac{\boldsymbol{x}^*+\boldsymbol{y}^*}{2}) < f(\boldsymbol{x}^*)$, $\boldsymbol{x}^*$ can't be optimal since the point $\frac{\boldsymbol{x}^*+\boldsymbol{y}^*}{2}$ is better. This contradiction proves that there is only one solution. ■

The simplest type of convex optimization is *linear program* (LP), where $f_0, f_1, \dots, f_m, h_1, \dots, h_k$ are all affine functions.

Definition 1 (Linear program)**.** Linear program (LP) can be written as:

$$\begin{aligned} \min_{\boldsymbol{x}\in\mathbb{R}^n} \quad & \boldsymbol{a}_0^{\mathrm{T}}\boldsymbol{x} - b_0, \\ \text{s.t.} \quad & \boldsymbol{a}_i^{\mathrm{T}}\boldsymbol{x} - b_i = 0,\ i=1,\dots,m, \\ & \boldsymbol{c}_j^{\mathrm{T}}\boldsymbol{x} - d_j \le 0,\ j=1,\dots,k, \end{aligned}$$

where $\boldsymbol{a}_0, \boldsymbol{a}_i, \boldsymbol{c}_j \in \mathbb{R}^n$ and $b_0, b_i, d_j \in \mathbb{R}$.
Since $-b_0$ is a constant and doesn't affect the solution, it can be dropped. Define:

$$\mathbf{A} = \begin{bmatrix} \boldsymbol{a}_1^{\mathrm{T}} \\ \vdots \\ \boldsymbol{a}_m^{\mathrm{T}} \end{bmatrix}, \quad \boldsymbol{b} = \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} \boldsymbol{c}_1^{\mathrm{T}} \\ \vdots \\ \boldsymbol{c}_k^{\mathrm{T}} \end{bmatrix}, \quad \boldsymbol{d} = \begin{bmatrix} d_1 \\ \vdots \\ d_k \end{bmatrix},$$

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Then LP can be written as

$$\begin{aligned} \min_{\boldsymbol{x}\in\mathbb{R}^n} \quad & \boldsymbol{a}_0^{\mathrm{T}}\boldsymbol{x}, \\ \text{s.t.} \quad & \mathbf{A}\boldsymbol{x} = \boldsymbol{b}, \\ & \mathbf{C}\boldsymbol{x} \leq \boldsymbol{d}. \end{aligned}$$

(The last vector inequality holds element-wise.)
The feasible set is $\mathcal{S} = \{\boldsymbol{x} \in \mathbb{R}^n \mid \mathbf{A}\boldsymbol{x} = \boldsymbol{b},\ \mathbf{C}\boldsymbol{x} \leq \boldsymbol{d}\}$.

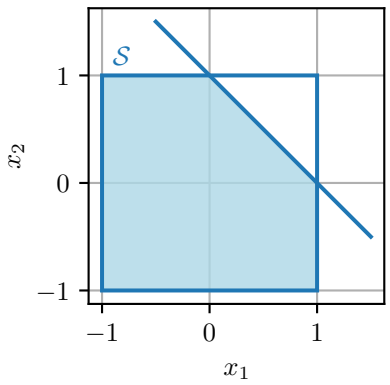
The feasible set $\mathcal{S}$ is the intersection of a finite number of half-spaces and hyperplanes, which is called a *polyhedron*:

$$\mathcal{S} = \left(\bigcap_{i=1}^{m} \underbrace{\{\boldsymbol{x} \in \mathbb{R}^n \mid \boldsymbol{a}_i^{\mathrm{T}}\boldsymbol{x} = b_i\}}_{\text{hyperplane}} \right) \cap \left(\bigcap_{j=1}^{k} \underbrace{\{\boldsymbol{x} \in \mathbb{R}^n \mid \boldsymbol{c}_j^{\mathrm{T}}\boldsymbol{x} \leq d_j\}}_{\text{half-space}} \right).$$

Example 1. Consider

$$\begin{aligned} \min_{\boldsymbol{x}\in\mathbb{R}^2} \quad & x_1 - x_2, \\ \text{s.t.} \quad & -1 \leq x_1 \leq 1, \\ & -1 \leq x_2 \leq 1, \\ & x_1 + x_2 \leq 1. \end{aligned}$$

The feasible set $\mathcal{S}$ is the intersection of five half-spaces:



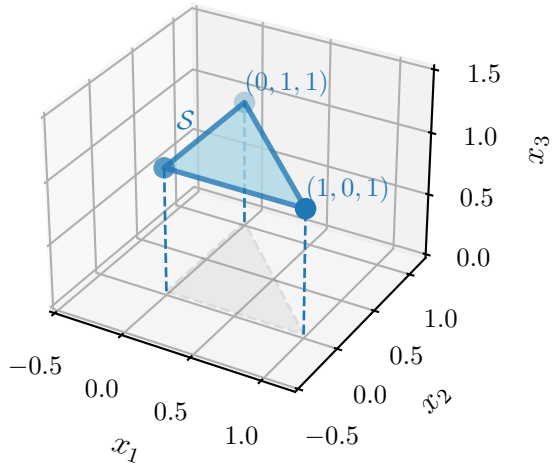
■

Example 2. Consider

$$\begin{aligned} \min_{\boldsymbol{x}\in\mathbb{R}^3} \quad & x_1 + x_2 - x_3, \\ \text{s.t.} \quad & x_1 + x_2 \leq 1, \\ & 0 \leq x_1, \\ & 0 \leq x_2, \\ & x_3 = 1. \end{aligned}$$

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The feasible set $\mathcal{S}$ is the intersection of three half-spaces and one hyperplane, which forms a lifted triangle:

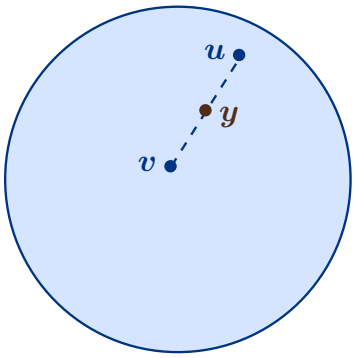


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Definition 2 (Extreme point)**.** Given a convex set $\mathcal{S}$, a point $\boldsymbol{y} \in \mathcal{S}$ is called an *extreme point* if there do not exist two other points $\boldsymbol{u}, \boldsymbol{v} \in \mathcal{S}$, such that $\boldsymbol{y}$ is a convex combination of $\boldsymbol{u}$ and $\boldsymbol{v}$ (i.e., $\boldsymbol{y} = \alpha\boldsymbol{u} + (1-\alpha)\boldsymbol{v}$ for some $0 < \alpha < 1$).

Example 3 (Extreme point)**.** Consider $\mathcal{S} = \{\boldsymbol{x} \in \mathbb{R}^2 \mid x_1^2 + x_2^2 \leq 1\}$.

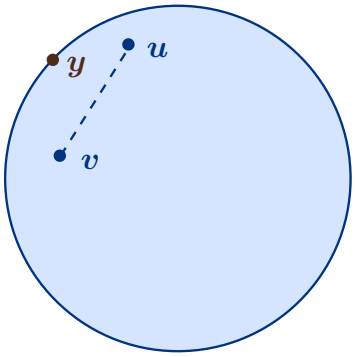
- Consider an interior point $\boldsymbol{y}$:



  We can always find $\boldsymbol{u}, \boldsymbol{v} \in \mathcal{S}$ such that $\boldsymbol{y}$ is on the segment from $\boldsymbol{u}$ to $\boldsymbol{v}$. Therefore, $\operatorname{int}\mathcal{S}$ has no extreme point.

- Now pick $\boldsymbol{y} \in \partial\mathcal{S}$ (boundary):

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Then such $\boldsymbol{u}$ and $\boldsymbol{v}$ don't exist. In this example, all boundary points are extreme points (but generally this isn't true).

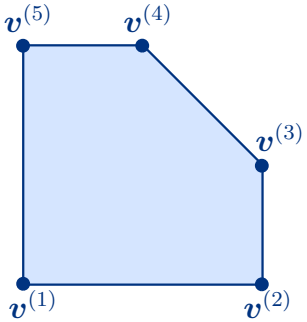
■

If $\mathcal{S}$ is a polyhedron, the extreme points are called *vertices*.

Example 4 (Vertices)**.** Consider the following set:

$$\begin{aligned} -1 \leq x_1 \leq 1, \\ -1 \leq x_2 \leq 1, \\ x_1 + x_2 \leq 1. \end{aligned}$$

There are 5 vertices $\boldsymbol{v}^{(1)}, \ldots, \boldsymbol{v}^{(5)}$:



Note: There are no other extreme points. For example, consider a point $\boldsymbol{y}$ on the segment between $\boldsymbol{v}^{(1)}$ and $\boldsymbol{v}^{(5)}$:



Any $\boldsymbol{y}$ on this boundary line is a convex combination of $\boldsymbol{v}^{(1)}$ and $\boldsymbol{v}^{(5)}$, so it's not an extreme point. ■

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Consider LP:

$$\begin{aligned} \min \quad & \boldsymbol{a}_0^{\mathrm{T}}\boldsymbol{x}, \\ \text{s.t.} \quad & \mathbf{A}\boldsymbol{x} = \boldsymbol{b}, \\ & \mathbf{C}\boldsymbol{x} \leq \boldsymbol{d}. \end{aligned}$$

Let the optimal objective value be denoted as p^*. There're three scenarios:

(1) $\boldsymbol{x}^*$ exists, then p^* is finite. Example:

$$\begin{aligned} \min \quad & x, \\ \text{s.t.} \quad & 2x \geq 1 \end{aligned} \quad \Rightarrow \quad \begin{cases} x^* = 1/2, \\ p^* = 1/2. \end{cases}$$

(2) The problem is unbounded from below, then we say $p^* = -\infty$. Example:

$$\begin{aligned} \min \quad & x_1 + x_2, \\ \text{s.t.} \quad & x_1 - x_2 \leq 1. \end{aligned}$$

Pick $x_2 = 0$ and $x_1 \to -\infty$, then the objective function goes to $-\infty$. Thus the problem is unbounded from below and $p^* = -\infty$.

Note that in this case, we don't say $\boldsymbol{x}^* = [-\infty, 0]^{\mathrm{T}}$; instead, we say that $\boldsymbol{x}^*$ does not exist.

(3) The problem is infeasible, i.e., the feasible set is empty. Example:

$$\begin{aligned} \min \quad & x, \\ \text{s.t.} \quad & x \geq 0, \\ & x \leq -1. \end{aligned}$$

In this case, we say $p^* = +\infty$.

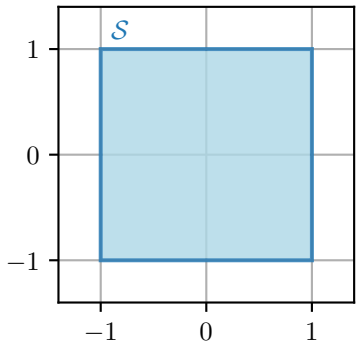
Theorem 2. *Assume LP has a solution. Then, at least one of the vertices of its feasible set is a solution.*

To understand this, consider LP:

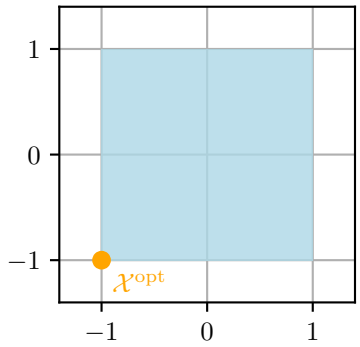
$$\begin{aligned} \min_{\boldsymbol{x} \in \mathbb{R}^2} \quad & \beta_1 x_1 + \beta_2 x_2, \\ \text{s.t.} \quad & -1 \leq x_1 \leq 1, \\ & -1 \leq x_2 \leq 1, \end{aligned}$$

where β_1 and β_2 are constants. The feasible set has 4 vertices, $[\pm 1, \pm 1]^{\mathrm{T}}$:

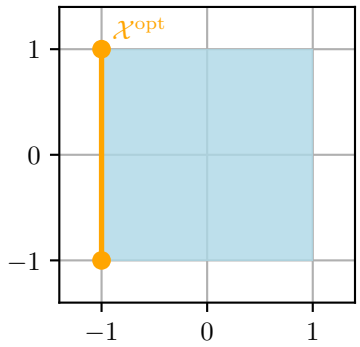
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(a) If $\beta_1 > 0$ and $\beta_2 > 0$, then $[-1, -1]^{\mathrm{T}}$ is the only solution and it is a vertex.

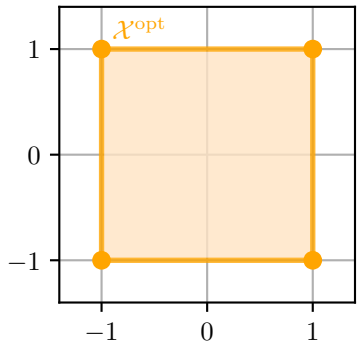


(b) If $\beta_1 > 0$ and $\beta_2 = 0$, then $\mathcal{X}^{\mathrm{opt}}$ is a line segment, which includes two vertices:



(c) If $\beta_1 = 0$ and $\beta_2 = 0$, then the entire box is $\mathcal{X}^{\mathrm{opt}}$ and it includes four vertices:

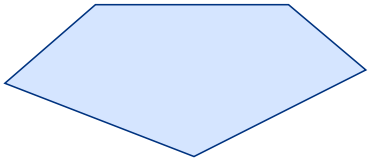
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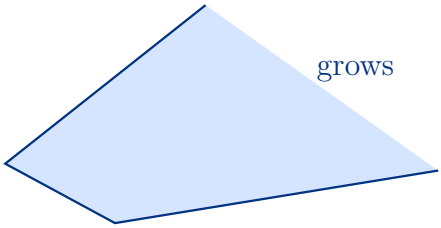
(We can continue to study other scenarios for β_1, β_2 such as $\beta_1, \beta_2 < 0$.)
From these cases, we observe that while the solutions may not be unique, they always include a vertex.

Proof. There are two scenarios:

(1) $\mathcal{S}$ is bounded:



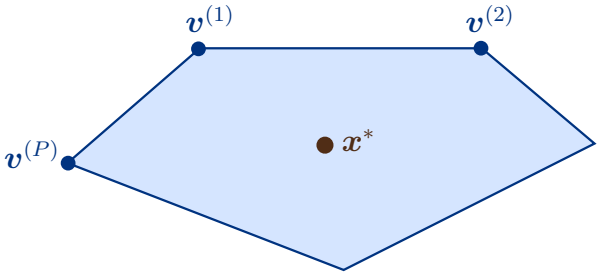
(2) $\mathcal{S}$ is unbounded (e.g., $\{\boldsymbol{x} \in \mathbb{R}^2 \mid x_1 \geq 0,\ x_2 \geq 0\}$):



Proof in case (2) is a bit involved, so we only prove case (1) here.
Let the vertices of the feasible set $\mathcal{S}$ be denoted as $\boldsymbol{v}^{(1)}, \ldots, \boldsymbol{v}^{(P)}$, where P is the number of vertices. Let $\boldsymbol{x}^*$ be an arbitrary solution. By contradiction, assume none of the vertices is a solution, so

$$\boldsymbol{a}_0^{\mathrm{T}} \boldsymbol{v}^{(i)} > \boldsymbol{a}_0^{\mathrm{T}} \boldsymbol{x}^*, \quad \forall i \in \{1, \ldots, P\}.$$

(This means that all vertices are worse than $\boldsymbol{x}^*$.)



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There is a lemma that says every point in a polyhedron can be written as a convex combination of its vertices. Therefore, there exists $\alpha_1, \ldots, \alpha_P \geq 0$ such that $\alpha_1 + \cdots + \alpha_P = 1$ and

$$\boldsymbol{x}^* = \alpha_1 \boldsymbol{v}^{(1)} + \alpha_2 \boldsymbol{v}^{(2)} + \cdots + \alpha_P \boldsymbol{v}^{(P)}.$$

Multiply both sides by $\boldsymbol{a}_0^{\mathrm{T}}$, we get

$$\begin{aligned}
\boldsymbol{a}_0^{\mathrm{T}} \boldsymbol{x}^* &= \alpha_1 \boldsymbol{a}_0^{\mathrm{T}} \boldsymbol{v}^{(1)} + \alpha_2 \boldsymbol{a}_0^{\mathrm{T}} \boldsymbol{v}^{(2)} + \cdots + \alpha_P \boldsymbol{a}_0^{\mathrm{T}} \boldsymbol{v}^{(P)} \\
&> \alpha_1 \boldsymbol{a}_0^{\mathrm{T}} \boldsymbol{x}^* + \alpha_2 \boldsymbol{a}_0^{\mathrm{T}} \boldsymbol{x}^* + \cdots + \alpha_P \boldsymbol{a}_0^{\mathrm{T}} \boldsymbol{x}^* \\
&= (\alpha_1 + \cdots + \alpha_P) \boldsymbol{a}_0^{\mathrm{T}} \boldsymbol{x}^* \\
&= \boldsymbol{a}_0^{\mathrm{T}} \boldsymbol{x}^*,
\end{aligned}$$

which is a contradiction. Therefore, Theorem 2 holds. ■

Theorem 2 can be used to design an algorithm for solving LP. To explain the idea, consider

$$\begin{aligned}
\min_{\boldsymbol{x} \in \mathbb{R}^2} \quad & \beta_1 x_1 + \beta_2 x_2, \\
\text{s.t.} \quad & -1 \leq x_1 \leq 1, \\
& -1 \leq x_2 \leq 1.
\end{aligned}$$

- To solve this, we can obtain all the vertices, i.e., $[\pm 1, \pm 1]^{\mathrm{T}}$, then check which one is better:

$$\min \left([\beta_1, \beta_2] \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \ [\beta_1, \beta_2] \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \ [\beta_1, \beta_2] \begin{bmatrix} -1 \\ 1 \end{bmatrix}, \ [\beta_1, \beta_2] \begin{bmatrix} -1 \\ -1 \end{bmatrix} \right).$$

- The vertex that leads to the minimum objective value is a solution, if a solution exists.
- Note: if the feasible set is bounded, then the solution must exist.
- Issue with this idea: how to find vertices in general?
- First we put a given LP into a standard LP via two operations:

  (1) Convert inequalities to equalities. For example, for $\boldsymbol{c}_1^{\mathrm{T}} \boldsymbol{x} \leq d_1$, we define a new variable s_1 (called a *slack variable*), and replace the inequality with $\boldsymbol{c}_1^{\mathrm{T}} \boldsymbol{x} + s_1 = d_1$ and $s_1 \geq 0$.

  (2) Make all variables non-negative. For variable x_i, we define two new variables x_i^+, x_i^-, and replace x_i with $x_i^+ - x_i^-$ with the constraints $x_i^+, x_i^- \geq 0$.
  The idea is that each arbitrary number can be written as the difference of two non-negative numbers. For example, $3 = 3 - 0 = 4 - 1$, $-5 = 0 - 5 = 1 - 6$ (note that x_i^+, x_i^- are not unique).

Example 5. Consider LP:

$$\begin{aligned}
\min_{x_1, x_2} \quad & x_1 - 4x_2, \\
\text{s.t.} \quad & x_1 + 5x_2 \leq 1, \\
& x_1 \geq 0.
\end{aligned}$$

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We can convert it into standard LP as follows:

$$\begin{aligned} \min_{x_1, x_2^+, x_2^-, s_1} \quad & x_1 - 4(x_2^+ - x_2^-), \\ \text{s.t.} \quad & x_1 + 5(x_2^+ - x_2^-) + s_1 = 1, \\ & x_1 \geq 0, \\ & x_2^+ \geq 0, \\ & x_2^- \geq 0, \\ & s_1 \geq 0. \end{aligned}$$

(Note: x_1 was already non-negative, so operation (2) is not needed for x_1.) ■

Every standard LP can be written as

$$\begin{aligned} \min_{\boldsymbol{x} \in \mathbb{R}^n} \quad & \boldsymbol{a}_0^{\mathrm{T}} \boldsymbol{x}, \\ \text{s.t.} \quad & \mathbf{A}\boldsymbol{x} = \boldsymbol{b}, \\ & \boldsymbol{x} \geq \mathbf{0}. \end{aligned}$$

(Note: $\boldsymbol{x}$ here is the vector of *all* variables and it includes slack variables s_i and x_i^+, x_i^-.)

- Assume $\mathbf{A}$ has full row rank. If this is not true, then some constraints can be eliminated. Example:

$$\begin{aligned} x_1 + x_2 + x_3 &= 1, \\ 2x_1 + 3x_2 + 4x_3 &= 2, \\ 3x_1 + 4x_2 + 5x_3 &= 3. \end{aligned}$$

  The third constraint is the sum of the first two constraints, so it can be eliminated. We eliminate redundant constraints until the new matrix $\mathbf{A}$ has full row rank.

- Assume $\mathbf{A} \in \mathbb{R}^{m \times n}$. Since it has full row rank, $m \leq n$.

- Procedure to find vertices of standard LP:

  (1) Select m arbitrary indices from the set $\{1, \ldots, n\}$ and call them $i_1, \ldots, i_m$.

  (2) Let $\boldsymbol{a}^{(i_1)}, \ldots, \boldsymbol{a}^{(i_m)}$ denote columns $i_1, \ldots, i_m$ of $\mathbf{A}$.

  (3) If $\boldsymbol{a}^{(i_1)}, \ldots, \boldsymbol{a}^{(i_m)}$ are linearly independent, then continue to the next step (otherwise stop).

  (4) Find the unique vector $\boldsymbol{y} \in \mathbb{R}^n$ such that $\mathbf{A}\boldsymbol{y} = \boldsymbol{b}$ and

$$y_j = 0, \quad \forall j \in \{1, \ldots, n\} \backslash \{i_1, \ldots, i_m\}.$$

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Note:

$$\mathbf{A}\boldsymbol{y} = \boldsymbol{b} \;\Leftrightarrow\; \begin{bmatrix}\boldsymbol{a}^{(i_1)} & \cdots & \boldsymbol{a}^{(i_m)}\end{bmatrix} \begin{bmatrix} y_{i_1} \\ \vdots \\ y_{i_m}\end{bmatrix} = \boldsymbol{b}$$

$$\Leftrightarrow \begin{bmatrix} y_{i_1} \\ \vdots \\ y_{i_m}\end{bmatrix} = \underbrace{\begin{bmatrix}\boldsymbol{a}^{(i_1)} & \cdots & \boldsymbol{a}^{(i_m)}\end{bmatrix}}_{\text{invertible due to step (3)}}{}^{-1}\boldsymbol{b}.$$

(5) If $\boldsymbol{y} \geq \mathbf{0}$, then $\boldsymbol{y}$ is a vertex.

Theorem 3. *Repeating steps (1)–(5) for all possible indices $\{i_1, \ldots, i_m\}$ will find all vertices.*

Example 6. Consider

$$\begin{aligned} \min_{\boldsymbol{x}\in\mathbb{R}^4} \quad & \boldsymbol{a}_0^{\mathrm{T}}\boldsymbol{x}, \\ \text{s.t.} \quad & \begin{bmatrix} 1 & 0 & -1 & 2 \\ 1 & 1 & -2 & 2\end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4\end{bmatrix} = \begin{bmatrix} 1 \\ 2\end{bmatrix}, \\ & \boldsymbol{x} \geq \mathbf{0}. \end{aligned}$$

- Select $i_1 = 1, i_2 = 2$, then $y_3 = 0, y_4 = 0$ and

$$\begin{bmatrix} 1 & 0 & -1 & 2 \\ 1 & 1 & -2 & 2\end{bmatrix}\begin{bmatrix} y_1 \\ y_2 \\ 0 \\ 0\end{bmatrix} = \begin{bmatrix} 1 \\ 2\end{bmatrix} \;\Rightarrow\; \begin{bmatrix} y_1 \\ y_2\end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 1\end{bmatrix}^{-1}\begin{bmatrix} 1 \\ 2\end{bmatrix} = \begin{bmatrix} 1 \\ 1\end{bmatrix}.$$

  Since $[1, 1, 0, 0]^{\mathrm{T}} \geq \mathbf{0}$, it is a vertex.

- Select $i_1 = 1, i_2 = 3$, then $y_2 = 0, y_4 = 0$ and

$$\begin{bmatrix} 1 & 0 & -1 & 2 \\ 1 & 1 & -2 & 2\end{bmatrix}\begin{bmatrix} y_1 \\ 0 \\ y_3 \\ 0\end{bmatrix} = \begin{bmatrix} 1 \\ 2\end{bmatrix} \;\Rightarrow\; \begin{bmatrix} y_1 \\ y_3\end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 1 & -2\end{bmatrix}^{-1}\begin{bmatrix} 1 \\ 2\end{bmatrix} = \begin{bmatrix} 0 \\ -1\end{bmatrix}.$$

  Since $\boldsymbol{y} \ngeq \mathbf{0}$, it is not a vertex.

- Select $i_1 = 1, i_2 = 4$, then the corresponding columns $\boldsymbol{a}^{(1)} = [1, 1]^{\mathrm{T}}$ and $\boldsymbol{a}^{(4)} = [2, 2]^{\mathrm{T}}$ are linearly dependent. Thus it can't produce a vertex.

■

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The number of vertices is less than or equal to

$$\binom{n}{m} = \frac{n!}{m!(n-m)!}.$$

Note: when solving

$$\begin{bmatrix} \boldsymbol{a}^{(i_1)} & \cdots & \boldsymbol{a}^{(i_m)} \end{bmatrix} \begin{bmatrix} y_{i_1} \\ \vdots \\ y_{i_m} \end{bmatrix} = \boldsymbol{b},$$

it could happen that at least one of $y_{i_1}, \ldots, y_{i_m}$ is zero. This is called the *degenerate* case. If a standard LP is non-degenerate, each vertex has $n - m$ zeros and m non-zero elements.

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